

CLINIC OPERATIONS + RESPONSIBLE AI

Arrive ready. Start care sooner.

Team	ClinicPass
Build status	Runnable MVP; live AGNES vision endpoint verified

Executive summary

ClinicPass moves administrative pre-registration ahead of arrival for scheduled and walk-in primary-care visits. Patients use a clearly simulated Singpass/MyInfo handoff or manual entry, state their visit reason and document need, then photograph, upload, or select a watermarked synthetic sample. PDFs become page images and photos are normalized; AGNES receives private visual inputs together with local OCR evidence and extracts administrative facts through constrained tool calls. Six transparent rules evaluate identity details, validity, clinic restrictions, organization/package codes, supporting documents, and billing completeness. A patient who declares no documents or uncertainty can still reach staff review without a false failure. Submission issues a live queue number; clinic actions update the patient's destination in real time. Clinic staff inspect grounded evidence, resolve exceptions, and make the final administrative decision. Identity, e-card, and originals are always verified in person. The runnable Docker prototype includes responsive patient and clinic/manager Web UIs, PostgreSQL, an asynchronous worker, five tablet-ready fixtures, audit events, Mailpit notifications, mock Clinic Assist export, and a documented Microsoft Copilot Studio port. Fixture mode supports rehearsals; live AGNES errors fail visibly rather than silently substituting results. ClinicPass does not make clinical decisions or guarantee coverage.

The problem

The source specification identifies a 13-20 minute administrative block before queueing that is addressable, not the entire 23-32 minute visit-registration journey. Documents often arrive late, formats vary, package knowledge stays tacit, staff retype data, and exceptions lack a consistent workflow.

Late capture Eligibility documents surface at the counter, leaving no time to resolve missing pages or expired dates.	Non-standard evidence Referrals, vouchers, and authorizations use inconsistent wording and layouts.
Repeated entry Patients and staff re-enter the same identifiers and package details across disconnected steps.	Exception overload Tacit package rules make every case feel manual instead of highlighting only what needs judgement.

VALUE PROPOSITION	Prepare administrative evidence before arrival; let transparent rules find exceptions; keep staff in control.
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Solution and user journey



Patient surface

Simulated Singpass/MyInfo or manual entry, visit reason and document declaration, phone camera/file upload, five tablet-ready synthetic forms, immediate parsing, patient-safe summary, live queue/room updates, and documentless review path.

Clinic surface

Prioritized queue, patient declarations, field/evidence review, information request, approval with recorded override, on-site check-in, live destination updates, and guarded Clinic Assist export. Managers see references, metrics, and audit.

Clean journey

- Patient submits before arrival; OCR and AGNES return evidence-linked administrative facts.
- Six deterministic checks return PASS or REVIEW; staff approve after inspecting evidence.
- At arrival, staff physically check identity, e-card, and originals before export.

Exception journey

- Expired, conflicting, missing, or unreadable evidence is presented as a focused exception.
- Staff draft and confirm a request; Mailpit simulates delivery; the case moves to NEEDS_ACTION.
- Any approval with a failing check requires an explicit override and reason in the audit trail.

NON-NEGOTIABLE
BOUNDARY

ClinicPass never remotely verifies identity or e-card, makes clinical decisions, or guarantees coverage. Final administrative readiness remains a staff decision; originals are checked on site.

What is innovative

The novelty is not autonomous eligibility. It is a traceable division of labor: local layout evidence, constrained model extraction, transparent administrative rules, exception-first human review, and a controlled handoff. The same domain actions are reusable by Web UI, future Android, and Copilot Studio without moving decision authority into a chatbot.

Architecture, feasibility, and governance

Interfaces	Patient Web UI Clinic assistant + manager Web UI Future Android Future Copilot Studio
Gateway + API	Caddy Next.js 16 / React / TypeScript FastAPI / Pydantic secure cookie sessions + role and clinic claims
Processing	PostgreSQL job queue Python worker normalized PDF/photo images Tesseract OCR evidence AGNES vision + forced tool calls
Integrations	Document volume Mailpit mock Clinic Assist HTTP adapter future managed storage and production connector

AGNES vision + OCR Up to six PDF pages or normalized photos are sent as private data URLs beside OCR evidence. Two schema-constrained tool calls classify and extract. Pydantic validates shape; invented evidence IDs are discarded; failures remain visible.	Deterministic code Uploads process during draft entry. Rules decide readiness only after submission. A no/unsure document declaration becomes staff REVIEW; a failing check needs explicit override. Every consequential event enters the audit trail.
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Copilot Studio controlled action map

Seven documented REST actions: search cases; get summary; explain exceptions; locate evidence; draft patient request; submit confirmed request; record confirmed decision. The OpenAPI v2 JSON is ready to import. The production port uses Microsoft Entra OAuth/OBO so role and clinic authorization come from signed-in staff claims, not model text.

Governance and security

- Synthetic data only; API keys stay server-side; opaque sessions are hashed at rest; uploads are size/type limited.
- Evidence snippets constrain model context and reduce prompt-injection surface; model output never invokes approval directly.
- Mock Clinic Assist proves the adapter contract. No NEHR write is in scope; any future clinical-record connection requires separate consent, minimization, governance, and interface review.

Feasible build path

Completed: Docker stack, both Web surfaces, simulated identity handoff, generic camera/file intake, draft processing, AGNES vision adapter, five synthetic artifacts, migrations, six rules, mock integrations, Copilot contract, public exporter, and tests. Production dependencies remain: official labelled fixtures, Clinic Assist contract, tenant identity, retention policy, and pricing.

Operational impact, cost, scalability, and ask

Measured prototype evidence

21	5	390 px	0
API tests passed	Tablet demo forms	Mobile acceptance width	Silent AI fallbacks

The live AGNES endpoint read a synthetic medical-chit image from a private data URL. Automated tests cover intake, uploads, retries, image normalization, evidence validation, and access boundaries. Static checks, the production build, Docker/browser journeys, and public-export checks pass.

Measurement plan - no unmeasured claim

With the official labelled fixture set, report field and evidence accuracy, false-pass rate, ready-before-arrival rate, duplicate-entry reduction, staff time, latency, and cost. The target is only the identified 13-20 minute administrative block; no impact claim is made yet.

Observed cost per case	OCR pages x OCR price/page + AGNES input/output tokens x tested price + messages + storage/retention + platform allocation
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Scale and configuration

Clinic, organization, package, and rule data are configuration, not prompt text. Stateless services and independent workers scale horizontally; PostgreSQL locking prevents duplicate jobs. Expansion follows only after organization-specific rules, governance, and contracts are validated.

Proposed pilot and ask

- Pilot one clinic with synthetic/shadow-mode records and a labelled document set; no production decision automation.
- Confirm official fixtures, pricing, Clinic Assist contract, tenant identity, organization rules, retention, privacy, security review, and operational ownership.
- Go/no-go: zero false passes in the agreed set, auditable evidence, staff acceptance, and measured net time/cost benefit.

APPENDIX: AI TOOLS USED	AGNES 2.0 Flash performs live classification and administrative extraction from synthetic document images. OpenAI Codex assisted implementation, testing, debugging, and documentation. All generated code, model outputs, and submission claims were reviewed and validated by the team. Microsoft Copilot Studio is documented as a future port and was not implemented.
PUBLIC REPOSITORY	https://github.com/HappyCool121/hackforhealth